

Jiachi ZHANG

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EDUCATION

South China University of Technology

09/2023 - 06/2027

Qualification: Bachelor of Engineering

Major: Information Engineering

Average Score: 84.52/100

Core Courses: Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France

07/2025

Core Courses: Français, théorie de l'information

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

10/2025 - Present

Position: *Visiting Research Intern*

- Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.
- Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.
- Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

RESEARCH EXPERIENCE

Efficient Medical Image Segmentation via Recursive Parameter Reuse

09/2025 - Present

- Studied medical image segmentation efficiency under extreme parameter scales, and adopted recursive parameter reuse to preserve segmentation fidelity with minimal parameter consumption.
- Built a multi-granularity recursive lightweight model family, enabling the 1.3K-parameter model to attain over 0.71 mean Dice and 21,009× parameter efficiency against U-Net (work under review at ACM Multimedia 2026).
- Designed a plug-and-play recursive refinement module with only ~0.00019M parameters and zero static computation, raising the Dice score by over 1.3 percentage points (work under review at ICONIP 2026).
- Boosted incremental parameter efficiency by four orders of magnitude over conventional segmentation refinement blocks, with negligible memory and latency overhead.

Deep Learning-based Glomerular Pathological Image Detection

04/2024 - 03/2025

- Designed color clustering and color transfer modules to mitigate staining variance and augment glomerular pathological image datasets.
- Constructed a YOLO11-based detection framework effectively and adopted Focal Loss to balance positive-and-negative samples in the glomerular detection task.
- Integrated EMA multi-scale attention to enhance feature fusion and suppress tissue noise; validated the method via ablation experiments, achieving 94% mAP that outperformed baseline models.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting System via SAITS and Time-Series LLMs

07/2025 - 08/2025

- Adopted the SAITS algorithm to conduct multivariate time-series imputation, and reconstructed high-fidelity continuous feature spaces under limited labeled data.
- Integrated TimeLLM and ChatTS, designed cross-modal alignment modules to capture temporal evolution trends and enhance prediction accuracy.

- Constructed a modular training and evaluation system, and developed high-precision interpretable time-series models, securing top 10 in the E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game

03/2026

- Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control and real-time score display.
- Designed HDMI display drivers and SDRAM caching logic, and implemented AABB-based collision detection for game objects.
- Used finite state machines to control seven game states and ensure synchronization between peripherals and display modules.
- Completed project validation and received the Excellent Course Design award for outstanding academic performance.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee

11/2024 - 11/2025

- Participated in the production and singing of the department song, and performed in the college band on various occasions.
- Managed and coordinated internal departmental affairs, and took charge of the college graduation gala as a key organizer for multiple times.

Study Representative, Class Committee

09/2023 - 10/2025

- Collected and distributed daily assignments, and completed the liaison work for academic affairs.
- Communicated actively between teachers and classmates, and released official university notifications in a timely manner.

HONORS & AWARDS

3rd Prize Scholarship , South China University of Technology	12/2025
Merit Student , South China University of Technology	12/2025
Advanced Individual in Student Work , South China University of Technology	09/2025
Active Participant in Student Work , South China University of Technology	09/2025
10th Place , Pazhou Algorithm Competition - E Fund Cup FinTech Challenge	08/2025
Outstanding League Cadre , South China University of Technology	05/2025

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)

Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado